

Siyu You

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EDUCATION

- **University of Electronic Science and Technology of China (UESTC)** Chengdu, China
B.Eng. in Electronic Information Engineering Sep. 2022 – Jun. 2026
GPA: 3.91/4.00 (89.12/100) Rank: 17/246
Major Coursework: Circuit Analysis, Digital Signal Processing, Signals and Systems, Embedded Processors, Digital Communication, Electronic Devices, Machine Learning and AI
- **University of Glasgow** Glasgow, United Kingdom
B.Eng. in Electronics and Electrical Engineering Sep. 2022 – Jun. 2026
Dual Degree Program with UESTC; Degree Classification: **First Class Honours**

PUBLICATIONS

- **Journal Papers**
 - **Siyu You**, Huayi Gou, Leilei Yang, Zhiliang Liu*, Mingjian Zuo, “Local Flaw Detection with Adaptive Pyramid Image Fusion Across Spatial Sampling Resolution for SWRs”, *IEEE Sensors Journal* (JCR Q1, IF=4.5).
 - Leilei Yang, Zhiliang Liu*, **Siyu You**, and Mingjian Zuo, “ILF Diagnosis for Steel Wire Ropes Based on Physical Information under Uniform Circular Array,” *NDT & E International* (JCR Q1, IF=4.5).
- **Conference Papers**
 - **Siyu You**, Leilei Yang, Zixu Kuang, Huayi Gou, Longlong Zhang, Zhiliang Liu*, “A Signal Matrix-Based Local Flaw Detection Framework for Steel Wire Ropes Using Convolutional Neural Networks”, *2025 International Conference on Mechatronics and Automation*, Oral Presentation.
 - Leilei Yang, **Siyu You**, Hailin Zhou, Zhiliang Liu*, “Weak LF Signal Detection Based on Multiscale Phase Spectrum Reconstruction for Steel Wire Ropes”, *IEEE PHM’Xi-an 2025* (**Best Paper Award**).
 - Boyuan Gu, Yanhui Yang, **Siyu You**, Haiyang Sun, Shisheng Guo*, “Improved VMD Based Remote Heartbeat Estimation Utilizing 60GHz mmWave Radar”, *2025 IEEE International Conference on Systems, Man, and Cybernetics (SMC 2025)*.

MANUSCRIPTS UNDER REVIEW

- **Siyu You**, Leilei Yang, Zhiliang Liu*, “MFLFormer: A Robust Physics-Informed Unsupervised Framework for Industrial Local Flaw Inspection of Steel Wire Ropes”, *IEEE Transactions on Industrial Informatics*, under revision.
- Shuaiqi Cheng, Jiahao Zhang, Boyuan Gu, **Siyu You**, Zexin Lu, Huiyu Zhou, Xuming Hu, “TamedFlow: Trajectory-Aware Distillation for Self-Speculative Diffusion Transformers”, *NeurIPS 2026*, under review.
- Leilei Yang, Hailin Zhou, **Siyu You**, Wu Wen, Fang Xie, Fengqiang Hu, Liming Yang, Zhiliang Liu, “Triaxial Magnetic Flux Leakage Data Fusion for Quantification of Local Flaws in Carrying–Hauling Ropes”, *NDT & E International*, under review.

WORK EXPERIENCE

- **Algorithm Engineer Intern** Xiaomi
Wearable Algorithms Team Mar. 2026 – Present
 - **VO₂ Max Estimation**: Developed wearable VO₂ max estimation algorithms for Xiaomi Band using physiological and motion signals, reducing estimation MAPE from 18% to 9% on internal validation data and outperforming competing wearable devices in real-world evaluation.
 - **Running Performance Prediction**: Built long-distance running performance prediction models with feature engineering, model training, error analysis, and robustness evaluation against reference measurements.

RESEARCH EXPERIENCE

- **WiFi-LLM: Domain-Specific Large Language Model for WiFi CSI Sensing** UESTC
Final-Year Project, Awarded as Outstanding Thesis (Top 10%) Sep. 2025 – Present
 - **Framework Design:** Developed a two-stage WiFi CSI sensing framework that combines masked reconstruction-based self-supervised pretraining with CSI-to-LLM alignment for signal-to-text human activity recognition.
 - **Model Development:** Designed a frequency-aware ResNet encoder and lightweight GRU temporal module to learn transferable temporal-frequency CSI representations, then projected the pretrained features into a frozen TinyLLaMA-1.1B embedding space using a lightweight MLP projector.
 - **Performance:** Achieved 92.40%, 96.21%, and 80.25% accuracy on UT-HAR, NTU-Fi, and WiFall, respectively, and validated the framework through baseline comparisons, ablation studies, few-shot transfer, feature visualization, efficiency analysis, and failure case analysis.
- **Quantification and Unsupervised Localization for Industrial Steel Wire Rope Defect** UESTC
Supervised by Prof. Zhiliang Liu Feb. 2025 – Present
 - **Physics-consistent Modeling:** Modeled multi-channel magnetic flux leakage signals as native signal matrices while preserving uniform circular sensor array topology to capture spatial channel correlations. Derived axial and circumferential diffusion patterns from magnetic dipole theory to construct a joint angular-depth localization model with Radial Basis Function-based quantification.
 - **Unsupervised Framework:** Designed a reconstruction-based anomaly detection architecture with a multi-channel Transformer encoder and CNN decoder. Introduced local-window self-attention and a physics-consistent reconstruction loss to enhance robustness and interpretability.
 - **Performance:** Achieved 99.4% F1-score under fully unsupervised settings with inference speed exceeding 100 FPS, localization error below 1 mm, and quantification error within one broken wire, supporting real-time industrial inspection.
- **Dual-task Learning for ECG-based Human Identity and Activity Recognition** UESTC
Supervised by Prof. Shisheng Guo Oct. 2024 – May 2025
 - **Model Design:** Designed a multi-scale residual convolutional shared backbone to extract ECG morphological patterns across temporal resolutions for joint identity and activity modeling.
 - **Feature Modulation:** Proposed a Subject-Adaptive Feature Modulation mechanism that leverages stable identity representations to calibrate activity features, mitigating cross-subject distribution shifts and improving generalization.
 - **Evaluation:** Achieved state-of-the-art performance on two public datasets and one in-house dataset, with real-time inference capability suitable for embedded deployment.
- **Adaptive Multi-scale Defect Detection for Steel Wire Ropes** UESTC
Supervised by Prof. Zhiliang Liu Oct. 2023 – Oct. 2024
 - **Method Development:** Proposed an Adaptive Spatial-resolution Target-Feature-Oriented method to address scale inconsistency caused by varying inspection speeds and sampling rates. Designed adaptive convolutional templates and integrated multi-scale pyramid weighted feature fusion to enhance local flaw responses.
 - **Performance:** Achieved 96.70% F1-score on real industrial datasets, reaching advanced performance in both detection accuracy and computational efficiency.

HONORS AND AWARDS

- **Scholarships:** Yanbao Scholarship (Top 5% nationwide, 2025); James Watt Innovation Scholarship (Top 5 in Glasgow College, 2025); First-Class Scholarship (Top 10%, awarded consecutively 2023, 2024, 2025).
- **Awards:** Outstanding Individual in Academic and Scientific Innovation, UESTC (Top 30 in UESTC, 2026); Team Leader, Top Ten Scientific Innovation Teams of the Year, Glasgow College, UESTC (2026); Gold Medal, International Exhibition of Inventions Geneva (2025); Silver Award, China International College Student Innovation Competition (2024); Top Ten Student Leaders of the Year, UESTC (2024).

LEADERSHIP ACTIVITIES

- **Peer Mentor**, Class of 2024 Freshmen, UESTC Aug. 2024 – Jun. 2025
 - Guided and supported incoming freshmen through their transition to university life, organized orientation activities, and acted as a liaison between students and faculty.
- **President**, Piano Association, UESTC Sep. 2023 – Aug. 2024
 - Managed association operations including budgeting and scheduling. Led a team of 10 members to organize musical events attracting over 1000 participants annually.

PROFESSIONAL SKILLS

- **Software**: Proficient in Ansys Maxwell, COMSOL, Vivado, LTspice, Origin
- **Programming**: Advanced in MATLAB, C, Python (PyTorch, TensorFlow), LaTeX; Basic in Verilog
- **Languages**: Chinese Mandarin (Native), English (IELTS: 7.5, GRE: 324)